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Dual down-set conductive terminals for externally mounted passive components

Abstract

In some examples, a semiconductor package comprises a first die pad having a first top surface and a first bottom surface opposing the first top surface and a second die pad in horizontal alignment with the first die pad, the second die pad having a second top surface and a second bottom surface opposing the second top surface, a gap separating the first and second die pads. The package comprises a semiconductor die electrically coupled to the first and second bottom surfaces and extending across the gap, the semiconductor die having a device side including circuitry formed therein, the device side facing away from the first and second die pads. The package comprises a first conductive terminal extending from a lateral surface of the first die pad and away from the first die pad, the first conductive terminal having multiple down-sets. The package comprises a second conductive terminal extending from a lateral surface of the second die pad and away from the second die pad, the second conductive terminal having multiple down-sets. The package comprises a third conductive terminal having fewer down-sets than the first and second conductive terminals and extending from below the first die pad and away from the first die pad. The package comprises a bond wire coupling the device side of the semiconductor die to the third conductive terminal. The package comprises a mold compound covering the semiconductor die and the bond wire, the first and second die pads exposed to an exterior of the mold compound through a top surface of the mold compound. The package comprises a passive electrical component external to the mold compound and having a first electrical contact coupled to the first die pad and having a second electrical contact coupled to the second die pad. The first, second, and third conductive terminals include distal ends that are exposed through a common side surface of the mold compound, the distal ends in horizontal alignment with each other.

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Background/Summary

BACKGROUND

(1) A semiconductor package may include a semiconductor die and a mold compound to cover the semiconductor die. The package may further include conductive terminals exposed to an exterior surface of the housing. The conductive terminals are coupled to the semiconductor die. The conductive terminals provide electrical pathways between circuitry on the semiconductor die and components (e.g., printed circuit boards) outside of the package.

SUMMARY

(2) In some examples, a semiconductor package comprises a first die pad having a first top surface

and a first bottom surface opposing the first top surface and a second die pad in horizontal alignment with the first die pad, the second die pad having a second top surface and a second bottom surface opposing the second top surface, a gap separating the first and second die pads. The package comprises a semiconductor die electrically coupled to the first and second bottom surfaces and extending across the gap, the semiconductor die having a device side including circuitry formed therein, the device side facing away from the first and second die pads. The package comprises a first conductive terminal extending from a lateral surface of the first die pad and away from the first die pad, the first conductive terminal having multiple down-sets. The package comprises a second conductive terminal extending from a lateral surface of the second die pad and away from the second die pad, the second conductive terminal having multiple down-sets. The package comprises a third conductive terminal having fewer down-sets than the first and second conductive terminals and extending from below the first die pad and away from the first die pad. The package comprises a bond wire coupling the device side of the semiconductor die to the third conductive terminal. The package comprises a mold compound covering the semiconductor die and the bond wire, the first and second die pads exposed to an exterior of the mold compound through a top surface of the mold compound. The package comprises a passive electrical component external to the mold compound and having a first electrical contact coupled to the first die pad and having a second electrical contact coupled to the second die pad. The first, second, and third conductive terminals include distal ends that are exposed through a common side surface of the mold compound, the distal ends in horizontal alignment with each other.

(3) In some examples, a method for manufacturing a semiconductor package comprises providing a metal structure including first and second die pads in horizontal alignment and separated by a gap, a first conductive terminal having a first end, and a second conductive terminal having a first end, the first end of the first conductive terminal extending closer to the first die pad in both the horizontal and vertical directions than does the first end of the second conductive terminal. The method also comprises coupling a semiconductor die to a bottom surface of the first die pad, wirebonding the semiconductor die to the metal structure, and covering the metal structure and the semiconductor die with a mold compound such that top surfaces of the first and second die pads are exposed through a top surface of the mold compound. The method also comprises cutting the mold compound and the metal structure such that second ends of the first and second conductive terminals are exposed through a side surface of the mold compound that is orthogonal to the top surface of the mold compound, the second ends of the first and second conductive terminals are horizontally aligned with each other and are vertically lower than the first and second die pads.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A is a perspective bottom view of die pads and dual down-set conductive terminals for a semiconductor package in accordance with various examples.

(2) FIG. **1**B is a perspective top-down view of die pads and dual down-set conductive terminals for a semiconductor package in accordance with various examples.

(3) FIG. **1**C is a top-down view of die pads and dual down-set conductive terminals for a semiconductor package in accordance with various examples.

(4) FIG. **1**D is a profile view of die pads and dual down-set conductive terminals for a semiconductor package in accordance with various examples.

(5) FIG. **1**E is a profile view of die pads and dual down-set conductive terminals for a semiconductor package in accordance with various examples.

(6) FIG. **1**F is a profile view of a die pad and dual down-set conductive terminals for a semiconductor package in accordance with various examples.

(7) FIG. **2**A is a perspective bottom view of dual down-set conductive terminals and of a semiconductor die coupled to die pads, in accordance with various examples.
(8) FIG. **2**B is a perspective bottom view of a semiconductor die coupled to die pads and to dual down-set conductive terminals, in accordance with various examples.
(9) FIG. **2**C is a bottom-up view of a semiconductor die coupled to die pads and to dual down-set conductive terminals, in accordance with various examples.
(10) FIG. **2**D is a perspective top view of a semiconductor die coupled to die pads and to dual down-set conductive terminals, in accordance with various examples.
(11) FIG. **2**E is a top-down view of a semiconductor die coupled to die pads and to dual down-set conductive terminals, in accordance with various examples.
(12) FIG. **2**F is a perspective bottom view of dual down-set conductive terminals and of a semiconductor die coupled to a die pad, in accordance with various examples.
(13) FIG. **2**G is a perspective bottom view of a semiconductor die coupled to a die pad and to dual down-set conductive terminals, in accordance with various examples.
(14) FIG. **2**H is a bottom-up view of a semiconductor die coupled to a die pad and to dual down-set conductive terminals, in accordance with various examples.
(15) FIG. **2**I is a perspective top view of a semiconductor die coupled to a die pad and to dual down-set conductive terminals, in accordance with various examples.
(16) FIG. **2**J is a top-down view of a semiconductor die coupled to a die pad and to dual down-set conductive terminals, in accordance with various examples.
(17) FIG. **3**A is a perspective view of a semiconductor package having dual down-set conductive terminals for externally mounted passive components, in accordance with various examples.
(18) FIG. **3**B is a top-down view of a semiconductor package having dual down-set conductive terminals for externally mounted passive components, in accordance with various examples.
(19) FIG. **3**C is a profile view of a semiconductor package having dual down-set conductive terminals for externally mounted passive components, in accordance with various examples.
(20) FIG. **3**D is a bottom-up view of a semiconductor package having dual down-set conductive terminals for externally mounted passive components, in accordance with various examples.
(21) FIG. **4**A is a perspective view of a semiconductor package having dual down-set conductive terminals for externally mounted passive components, in accordance with various examples.
(22) FIG. **4**B is a top-down view of a semiconductor package having dual down-set conductive terminals for externally mounted passive components, in accordance with various examples.
(23) FIG. **4**C is a profile view of a semiconductor package having dual down-set conductive terminals for externally mounted passive components, in accordance with various examples.
(24) FIG. **4**D is a bottom-up view of a semiconductor package having dual down-set conductive terminals for externally mounted passive components, in accordance with various examples.
(25) FIG. **5**A is a close-up view of a lateral surface of a semiconductor package having dual down-set conductive terminals for externally mounted passive components, in accordance with various examples.
(26) FIG. **5**B is a close-up view of a bottom surface of a semiconductor package having dual down-set conductive terminals for externally mounted passive components, in accordance with various examples.
(27) FIG. **6**A is a perspective view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.
(28) FIG. **6**B is a top-down view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.
(29) FIG. **6**C is a profile view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.
(30) FIG. **6**D is a profile view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.

(31) FIG. **6**E is a perspective view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.
(32) FIG. **6**F is a top-down view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.
(33) FIG. **6**G is a profile view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.
(34) FIG. **6**H is a profile view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.
(35) FIG. **6**I is a perspective view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.
(36) FIG. **6**J is a top-down view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.
(37) FIG. **6**K is a profile view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.
(38) FIG. **6**L is a profile view of a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.
(39) FIG. **7** is a flow diagram of a method for manufacturing a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples.

DETAILED DESCRIPTION

(40) Space within a semiconductor package is valuable. Many different components may compete for space within the package. Industry pressure favors continual reductions in package sizes, thus reducing the amount of available space even further. To conserve space within the package, some components may be relocated outside of the package. For example, passive components may be coupled to a top surface of a package. Such passive components may communicate with the semiconductor die within the package through metal interconnects that are exposed through the top surface of the package and that are coupled to the semiconductor die.
(41) Some types of packages, such as quad flat no lead (QFN) type packages, are relatively inexpensive to manufacture, but they are not as small as other, more expensive types of packages. A technique to reduce QFN package size—such as by relocating passive components to outside the package—without resorting to more expensive package types would be especially valuable, because such a technique would achieve smaller sizes comparable to those of more expensive packages without the increase in costs that accompanies such packages.
(42) This disclosure describes various examples of a QFN type semiconductor package that resolves the challenges described above. In examples, the semiconductor package comprises a first die pad having a first top surface and a first bottom surface opposing the first top surface and a second die pad in horizontal alignment with the first die pad. The second die pad has a second top surface and a second bottom surface opposing the second top surface. A gap separates the first and second die pads. The package also includes a semiconductor die coupled to the first and second bottom surfaces and extending across the gap. The semiconductor die has a device side including circuitry formed therein, with the device side facing away from the first and second die pads. The package further comprises a first conductive terminal extending from a lateral surface of the first die pad and away from the first die pad, the first conductive terminal having multiple down-sets. The package also includes a second conductive terminal extending from a lateral surface of the second die pad and away from the second die pad, the second conductive terminal having multiple down-sets. The package further comprises a third conductive terminal having a single down-set and extending from below the first die pad and away from the first die pad. The package includes a bond wire coupling the device side of the semiconductor die to the third conductive terminal. The package also includes a mold compound covering the semiconductor die and the bond wire, the first and second die pads exposed to an exterior of the mold compound through a top surface of the

mold compound. The package comprises a passive electrical component external to the mold compound and having a first electrical contact coupled to the first die pad and having a second electrical contact coupled to the second die pad. The first, second, and third conductive terminals include distal ends that are exposed through a common side surface of the mold compound. The distal ends in horizontal alignment with each other. This structure provides a technical solution to the technical problem of reducing QFN package size by relocating package components to a top surface of the package.

(43) FIGS. **1**A-**6**D are a process flow for manufacturing a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples. FIG. **7** is a flow diagram of a method **700** for manufacturing a semiconductor package having dual down-set conductive terminals and an externally mounted passive component, in accordance with various examples. Accordingly, the method **700** of FIG. **7** is described in parallel with the structures and process flow of FIGS. **1**A-**6**D.

(44) The method **700** begins with providing a metal structure including first and second die pads in horizontal alignment and separated by a gap (**702**). A first conductive terminal has a first end, and a second conductive terminal has a first end (**702**). The first end of the first conductive terminal extends closer to the first die pad in both the horizontal and vertical directions than does the first end of the second conductive terminal (**702**). FIG. **1**A is a perspective bottom view of die pads and dual down-set conductive terminals for a semiconductor package in accordance with various examples. Specifically, FIG. **1**A shows a die pad **102** and a die pad **104**. The die pads **102**, **104**, as well as the remaining components shown in FIG. **1**A, comprise any suitable conductive material, such as a metal (e.g., copper) or an alloy. The die pads **102**, **104** are horizontally aligned with each other, meaning that the die pads **102**, **104** at least partially overlap with each other in the horizontal direction. For example, bottom surfaces **101**, **103** of the die pads **102**, **104** may be flush with each other. A gap **105** separates the die pads **102** and **104** from each other. The gap **105** ranges from 50 microns to 2000 microns, with a smaller gap being disadvantageous because a mold compound, when applied, will form voids, and with a larger gap being disadvantageous because of an unnecessary increase in package size. FIG. **1**A also shows multiple conductive terminals **106**. Each conductive terminal **106** has a distal end **108** and a proximal end **110**. The ends **108** are distal to the die pads **102**, **104**, and the ends **110** are proximal to the die pads **102**, **104**. Each of the conductive terminals **106** includes a single down-set **112**. A down-set is a segment of a conductive terminal that extends vertically to couple first and second segments of the conductive terminal that are different horizontal distances from the die pads **102**, **104**. Although a down-set may extend vertically, it may also simultaneously extend both horizontally and vertically, as shown in FIG. **1**A. The proximal ends **110** do not couple to the die pads **102**, **104**. Rather, the proximal ends **110** are horizontally and vertically offset from the lateral surfaces **124** of the die pads **102**, **104**, as described further below.

(45) FIG. **1**A also shows multiple conductive terminals **114**. Each conductive terminal **114** has a distal end **116** and a proximal end **118**. The ends **116** are distal to the die pads **102**, **104**, and the ends **118** are proximal to the die pads **102**, **104**. Each of the conductive terminals **114** includes multiple down-sets **120**, **122**. The down set **120** is closer to the distal end **116**, and the down set **122** is closer to the proximal end **118**. Each of the proximal ends **118** is coupled to a lateral surface **124** of a die pad **102**, **104**. In some examples, the die pad **102** and the conductive terminals **114** to which it couples form a monolithic structure, meaning that the conductive terminals **114** and the die pad **102** are part of the same whole, rather than the conductive terminals **114** being formed separately and then being coupled to the die pad **102**. Similarly, in some examples, the die pad **104** and the conductive terminals **114** to which it couples form a monolithic structure, meaning that the conductive terminals **114** and the die pad **104** are part of the same whole, rather than the conductive terminals **114** being formed separately and then being coupled to the die pad **104**. However, for the sake of convenience, the conductive terminals **114** may be described herein as being coupled to the

die pads **102**, **104**. Accordingly, the proximal ends **118** are closer to the lateral surfaces **124** of the die pads **102** than the proximal ends **110**, and thus the proximal ends **118** are closer both horizontally and vertically to the lateral surfaces **124** than the proximal ends **110**.

(46) In some examples, each of the conductive terminals **106** has a single down-set (e.g., down-set **112**), and each of the conductive terminals **114** has dual down-sets (e.g., down-sets **120**, **122**). In some examples, each of the conductive terminals **106** has fewer down-sets than does each of the conductive terminals **114** (e.g., each of the conductive terminals has x down-sets, and each of the conductive terminals **106** has y down-sets, where y<x). In some examples, each of the conductive terminals **106** has the same number of down-sets as does each of the conductive terminals **114**, but at least one of the down-sets of the conductive terminals **114** extends farther in the vertical and horizontal directions than do the down-sets of the conductive terminals **106**.

(47) FIG. **1**B is a perspective top-down view of the die pads **102**, **104** and the conductive terminals **106**, **114** in accordance with various examples. The die pads **102**, **104** have top surfaces **128**, **130**, respectively. Further, the distal ends **108**, **116** of the conductive terminals **106**, **114** are vertically lower than the die pads **102**, **104**, and as numeral **126** indicates, the distal ends **108**, **116** of the conductive terminals **106**, **114** are in horizontal alignment with each other.

(48) FIG. **1**C is a top-down view of the die pads **102**, **104** and the conductive terminals **106**, **114**, in accordance with various examples. FIG. **1**D is a profile view of the die pads **102**, **104** and the conductive terminals **106**, **114**, in accordance with various examples. FIG. **1**E is a profile view of the die pads **102**, **104** and the conductive terminals **106**, **114**, in accordance with various examples. FIG. **1**F is a close-up, profile view of the die pad **102** and the conductive terminals **106**, **114**, in accordance with various examples. As shown, a vertical line represents the meeting of the proximal end **118** and the lateral surface **124**, but as explained above, the conductive terminal **114** and the die pad **102** may be part of the same whole, e.g., a monolithic structure. The proximal end **110** is vertically farther (e.g., lower) from the lateral surface **124** than is the proximal end **118**. Further, a gap **134** shows that the proximal end **110** is horizontally farther (e.g., more distal) from the lateral surface **124** than is the proximal end **118**. Stated another way, the proximal end **110** is not in horizontal alignment with the die pad **102** or the lateral surface **124**, and the proximal end **110** is not in vertical alignment with the die pad **102** or the lateral surface **124**. Vertical alignment means that two components at least partially overlap with each other in the vertical direction. The distance **132** shows the offset between the top surfaces of the conductive terminals **106**, **114** and has a range of 50 microns to 1000 microns, with a smaller distance being disadvantageous because of mold compound voiding challenges and mold compound cracking, and with a larger distance being disadvantageous because it presents significant lead frame manufacturing challenges. The gap **134** has a range of 50 microns to 2000 microns, with a smaller gap being disadvantageous because of mold compound voiding challenges, and with a larger gap being disadvantageous because it results in bond wire sweeps due to long bond wire loops and larger package sizes.

(49) The method **700** includes coupling a semiconductor die to a bottom surface of the first die pad (**704**). FIG. **2**A is a perspective bottom view of the conductive terminals **106**, **114** and of a semiconductor die coupled to the die pads **102**, **104**, in accordance with various examples. Specifically, FIG. **2**A shows a semiconductor die **200** coupled to the die pads **102**, **104**. In examples, the semiconductor die **200** is coupled to the bottom surfaces **101**, **103**. The semiconductor die **200** may be coupled to the bottom surfaces **101**, **103** using a die attach layer **202**. Alternatively, the semiconductor die **200** may be coupled to the bottom surfaces **101**, **103** using a suitable paste. In examples, because the semiconductor die **200** is coupled to the bottom surfaces **101** and **103**, the semiconductor die **200** spans the gap **105** and is partially visible from a top-down view of the structure of FIG. **2**A. The die attach layer **202**, if used, likewise spans the gap **105**. The semiconductor die **200** has a device side **204** in which circuitry is formed. The device side **204** faces away from the die pads **102**, **104**. Bond pads **206** are formed on the device side **204**. The bond pads **206** form interfaces at which bond wires or other suitable connectors may be coupled to

form electrical pathways with the circuit formed on the device side **204**.

(50) The method **700** includes wirebonding the semiconductor die to the metal structure (**706**). FIG. **2**B is a perspective bottom view of the structure of FIG. **2**A, in accordance with various examples. Wirebonds **208** are formed to couple the bond pads **206** to the conductive terminals **106**, **114** and/or to the die pads **102**, **104**. In some examples, the semiconductor die **200** may include electrical contacts on the device side **204** and on a non-device side opposing the device side **204** to facilitate coupling of the non-device side directly to the die pads **102**, **104**. In such examples, a paste may be useful to couple the semiconductor die **200** to the die pads **102**, **104** in a manner that facilitates coupling of electrical contacts on the non-device side of the die **200** to the die pads **102**, **104**. FIG. **2**C is a bottom-up view of the structure of FIG. **2**B, in accordance with various examples. FIG. **2**D is a perspective top view of the structure of FIG. **2**B, in accordance with various examples. FIG. **2**E is a top-down view of the structure of FIG. **2**B, in accordance with various examples. FIGS. **2**F-**2**J are identical to FIGS. **2**A-**2**E, respectively, except that the semiconductor die **200** is coupled to a single die pad instead of to multiple die pads.

(51) The method **700** also includes covering the metal structure and the semiconductor die with a mold compound such that the top surfaces of the first and second die pads are exposed through a top surface of the mold compound (**708**). FIG. **3**A is a perspective view of a semiconductor package having conductive terminals for externally mounted passive components, in accordance with various examples. Specifically, FIG. **3**A shows a semiconductor package **300** (e.g., a quad-flat no lead (QFN) package) including the structure of FIGS. **2**B-**2**E, covered by a mold compound **302**. The mold compound **302** has lateral surfaces **304** that are perpendicular to the top surface of the mold compound **302**. The top surfaces **128**, **130** of the die pads **102**, **104** are exposed through the top surface of the mold compound **302**. In examples, the top surfaces **128**, **130** are approximately flush with the top surface of the mold compound **302**. In examples, the distal ends **108** and **116** of the conductive terminals **106** and **114** are exposed through the lateral surfaces **304** of the mold compound **302**. In examples, the distal ends **108**, **116** are approximately flush with the lateral surfaces **304**, as shown. FIG. **3**B is a top-down view of the structure of FIG. **3**A, in accordance with various examples. FIG. **3**C is a profile view of the structure of FIG. **3**A, in accordance with various examples. FIG. **3**D is a bottom-up view of the structure of FIG. **3**A, in accordance with various examples. As FIG. **3**D shows, the distal ends **108**, **116** are exposed through a bottom surface of the mold compound **302** of the package **300**.

(52) The method **700** includes coupling a passive component to the top surfaces of the first and second die pads (**710**). FIG. **4**A is a perspective view of the structure of FIGS. **3**A-**3**D, except with the addition of printed solder members **306** and **308** on the top surfaces **128**, **130** of the die pads **102**, **104**, as shown. FIG. **4**B is a top-down view of the structure of FIG. **4**A, in accordance with various examples. FIG. **4**C is a profile view of the structure of FIG. **4**A, in accordance with various examples. FIG. **4**D is a bottom-up view of the structure of FIG. **4**A, in accordance with various examples. The method **700** may include cutting the mold compound and the metal structure such that second ends of the first and second conductive terminals are exposed through a side surface of the mold compound that is orthogonal to the top surface of the mold compound, the second ends of the first and second conductive terminals are horizontally aligned with each other and are vertically lower than the first and second die pads (**712**). The drawings assume that step **712** has already been performed, to facilitate clarity and ease of understanding. However, such package singulation may occur at any suitable time in the method **700**, including at the end of the method **700**, as FIG. **7** shows.

(53) FIG. **5**A is a close-up view of a lateral surface **304** of FIG. **3**A, in accordance with various examples. In some examples, multiple distal ends **108** are exposed through a lateral surface **304**, as described above. In some examples, the lateral surfaces **304** of the package **300** may include cavities **500** that facilitate the flow of solder during coupling to a printed circuit board. FIG. **5**B shows a bottom surface **310** of the package **300**, in which the cavity **500** extends to the bottom

surface **310**.

(54) FIG. **6**A is a perspective view of the semiconductor package **300** including an externally mounted passive component, in accordance with various examples. Specifically, FIG. **6**A shows a passive component **600** (e.g. inductor, capacitor, resistor) having a terminal **602** coupled to the printed solder member **306** and a terminal **604** coupled to the printed solder member **308**. FIG. **6**B is a top-down view of the structure of FIG. **6**A, in accordance with various examples. FIG. **6**C is a profile view of the structure of FIG. **6**A, in accordance with various examples. FIG. **6**D is a profile view of the structure of FIG. **6**A, in accordance with various examples.

(55) Although this description is provided primarily in the context of a quad flat no-lead (QFN) type package, other package types are contemplated. For example, FIG. **6**E shows a perspective view of a semiconductor package **610** that is a dual flat no-lead (DFN) type package. The structure of the DFN semiconductor package **610** of FIG. **6**E may be similar or the same as that for the QFN package of FIG. **1**A, but with conductive terminals omitted from two opposing sides, as shown. FIG. **6**F is a top-down view of the structure of FIG. **1**G, in accordance with various examples. FIG. **6**G is a profile view of the structure of FIG. **1**G, in accordance with various examples. FIG. **6**H is another profile view of the structure of FIG. **1**G, in accordance with various examples.

(56) Yet other package types are contemplated and included in the scope of this disclosure, including small outline transistor (SOT), small outline integrated circuit (SOIC), small outline package (SOP), dual inline package (DIP), and quad flat package (QFP) type packages. Other packages are also contemplated. FIG. **6**I shows a perspective view of a semiconductor package **620** having a gullwing style configuration, meaning that the example conductive terminals **106**, **114** of FIG. **6**I may be longer than the example conductive terminals **106**, **114** shown in other drawings and described above, such that the conductive terminals **106**, **114** extend beyond the lateral surfaces of the mold compound **302**. Because the conductive terminals **106**, **114** extend beyond the lateral surfaces of the mold compound **302** and extend vertically lower than a bottom surface of the mold compound **302**, the conductive terminals **106**, **114** are suitable for connecting (e.g., soldering) to a printed circuit board or other substrate. As described above, the conductive terminals **106** may have fewer down-sets than the conductive terminals **114**. FIG. **6**J is a bottom-up view of the structure of FIG. **6**I, in accordance with examples. FIG. **6**K is a profile view of the structure of FIG. **6**I, in accordance with examples. FIG. **6**L is a profile view of the structure of FIG. **6**I, in accordance with examples.

(57) In this description, the term “couple” may cover connections, communications or signal paths that enable a functional relationship consistent with this description. For example, if device A provides a signal to control device B to perform an action, then: (a) in a first example, device A is directly connected to device B; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, so device B is controlled by device A via the control signal provided by device A.

(58) A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or re-configurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

(59) In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within +/−10 percent of that parameter. Modifications are possible in the described examples, and other examples are possible within the scope of the claims.

Claims

1. A semiconductor package, comprising: a first die pad having a first top surface and a first bottom surface opposing the first top surface; a second die pad in horizontal alignment with the first die pad, the second die pad having a second top surface and a second bottom surface opposing the second top surface, a gap separating the first and second die pads; a semiconductor die electrically coupled to the first and second bottom surfaces and extending across the gap, the semiconductor die having a device side including circuitry formed therein, the device side facing away from the first and second die pads; a first conductive terminal extending from a lateral surface of the first die pad and away from the first die pad, the first conductive terminal having multiple down-sets; a second conductive terminal extending from a lateral surface of the second die pad and away from the second die pad, the second conductive terminal having multiple down-sets; a third conductive terminal having fewer down-sets than the first and second conductive terminals and extending from below the first die pad and away from the first die pad; a bond wire coupling the device side of the semiconductor die to the third conductive terminal; a mold compound covering the semiconductor die and the bond wire, the first and second die pads exposed to an exterior of the mold compound through a top surface of the mold compound; and a passive electrical component external to the mold compound and having a first electrical contact coupled to the first die pad and having a second electrical contact coupled to the second die pad, wherein the first, second, and third conductive terminals include distal ends that are exposed through a common side surface of the mold compound, the distal ends in horizontal alignment with each other.
2. The semiconductor package of claim 1, wherein the semiconductor package is a quad flat no lead (QFN) type package.
3. The semiconductor package of claim 1, wherein the semiconductor package is a dual flat no lead (DFN) type package.
4. The semiconductor package of claim 1, wherein the semiconductor package is a leaded package selected from the group consisting of small outline transistor (SOT) packages, small outline integrated circuit (SOIC) packages, small outline packages (SOP), dual inline packages (DIP), and quad flat package (QFP) type packages.
5. The semiconductor package of claim 1, further comprising a die attach layer coupling the semiconductor die to the first and second bottom surfaces.
6. The semiconductor package of claim 1, wherein the first and second die pads are electrically conductive.
7. The semiconductor package of claim 1, wherein the first and second top surfaces are approximately flush with the top surface of the mold compound.
8. The semiconductor package of claim 1, wherein the first, second, and third conductive terminals are exposed through a bottom surface of the mold compound that opposes the top surface of the mold compound.
9. The semiconductor package of claim 8, wherein the first, second, and third conductive terminals include cavities that are exposed through the common side surface of the mold compound and that are exposed through the bottom surface of the mold compound.
10. The semiconductor package of claim 1, wherein the bond wire is a first bond wire, and further comprising a second bond wire coupling the device side of the semiconductor die to the first die pad.
11. A quad flat no lead (QFN) semiconductor package, comprising: a first die pad; a second die pad horizontally aligned with the first die pad, the first and second die pads separated by a gap; a first conductive terminal having first and second opposing ends, the first end of the first conductive terminal coupled to the first die pad; a second conductive terminal having first and second opposing ends, the first end of the second conductive terminal located vertically lower than the first and second die pads and not vertically overlapping with the first die pad, the second ends of the first and second conductive terminals in horizontal alignment with each other; a semiconductor die

coupled to at least one of the first and second die pads; bond wires coupling the semiconductor die to the second conductive terminal and to at least one of the first and second die pads; and a mold compound covering the semiconductor die, the first and second die pads exposed through a top surface of the mold compound, the second ends of the first and second conductive terminals exposed through a side surface of the mold compound.

12. The QFN package of claim 11, wherein the first and second conductive terminals are exposed through a bottom surface of the mold compound.

13. The QFN package of claim 11, further comprising a passive electrical component external to the mold compound and coupled to the first and second die pads.

14. The QFN package of claim 13, wherein the passive electrical component is an inductor or a capacitor.

15. The QFN package of claim 11, wherein the first conductive terminal includes multiple down-sets.

16. The QFN package of claim 11, wherein the second conductive terminal includes a single down-set.

17. A method for manufacturing a semiconductor package, comprising: providing a metal structure including: first and second die pads in horizontal alignment and separated by a gap; a first conductive terminal having a first end; and a second conductive terminal having a first end, wherein the first end of the first conductive terminal extends closer to the first die pad in both the horizontal and vertical directions than does the first end of the second conductive terminal; coupling a semiconductor die to a bottom surface of the first die pad; wirebonding the semiconductor die to the metal structure; covering the metal structure and the semiconductor die with a mold compound such that top surfaces of the first and second die pads are exposed through a top surface of the mold compound; and cutting the mold compound and the metal structure such that second ends of the first and second conductive terminals are exposed through a side surface of the mold compound that is orthogonal to the top surface of the mold compound, the second ends of the first and second conductive terminals are horizontally aligned with each other and are vertically lower than the first and second die pads.

18. The method of claim 17, wherein the semiconductor package is a quad flat no lead (QFN) type package.

19. The method of claim 17, wherein the first conductive terminal includes multiple down-sets.

20. The method of claim 17, wherein the second conductive terminal includes only one down-set.

21. The method of claim 17, further comprising coupling a passive electrical component to the first and second die pads, the passive electrical component located outside of the mold compound.

22. The method of claim 17, further comprising coupling the semiconductor die to the bottom surface of the second die pad, the semiconductor die coupled to the bottom surfaces of the first and second die pads with a die attach layer, the die attach layer and the semiconductor die extending across the gap.